

Wednesday 10th June 2026

09:00 to 11:00

PHYSICS (5)**PHYSICAL SCIENCES: HALF SUBJECT PHYSICS (5)****ADVANCED AND QUANTUM OPTICS**

*Candidates offering this paper should attempt **all six** questions (four from Section A and two from Section B).*

The approximate number of marks allocated to each question or part of a question is indicated in the right margin. This paper contains 4 sides, including this coversheet, and is accompanied by a handbook giving values of constants and containing mathematical formulae which you may quote without proof.

STATIONERY REQUIREMENTS

2 × 8-page answer booklet
Rough workpad
Yellow master coversheet

SPECIAL REQUIREMENTS

Mathematical Formulae Handbook
Approved calculator allowed

You may not start to read the questions printed on the subsequent pages of this question paper until instructed that you may do so by the Invigilator.

SECTION A

Attempt **all four** questions in this Section. Answers should be concise and relevant formulae may be assumed without proof.

- 1 A collimated Gaussian beam of wavelength $\lambda = 1.0 \mu\text{m}$ and waist radius $W_0 = 0.50 \text{ mm}$ propagates in free space. State what is meant by the *Rayleigh range* of the beam, evaluate it for this beam, and hence estimate the beam diameter at a distance of 20 m from the waist. Comment briefly on why a larger waist gives a more collimated beam. [4]
- 2 A second-order nonlinear crystal is illuminated by a single monochromatic field of angular frequency ω . Explain physically why the nonlinear polarisation contains components at both 2ω and at zero frequency, and name the two corresponding optical processes. State the condition on the crystal symmetry that is required for these processes to occur, and explain why it is not satisfied in a centrosymmetric medium. [4]
- 3 A symmetric two-mirror optical resonator is constructed from two identical concave mirrors of radius of curvature R , separated by a distance d . Derive the stability (confinement) condition for this resonator in terms of d and R , and identify the values of $d/|R|$ that correspond to the planar, confocal and concentric configurations. State which of these is the most robust against mirror misalignment, and why. [4]
- 4 A single two-level atom is driven on resonance by a classical light field with resonant Rabi frequency Ω_0 . Sketch the population of the excited state as a function of time (i) in the absence of spontaneous emission, and (ii) when spontaneous emission at rate γ is included, with $\gamma \approx 0.2\Omega_0$. In each case state the behaviour at long times and explain physically the difference between the two cases. [4]

SECTION B

Attempt **both** questions from this Section.

5 This question concerns optical coherence and its use in interferometric imaging.

A quasi-monochromatic light source of mean frequency ν_0 emits a partially coherent plane wave whose complex degree of temporal coherence may be written $g(\tau) = g_a(\tau) \exp(j2\pi\nu_0\tau)$, where $g_a(\tau)$ is a slowly varying envelope.

- (a) Define the *coherence time* τ_c and the *coherence length* ℓ_c of the source, and state how τ_c is related to the spectral width $\Delta\nu_c$ of the light. [3]
- (b) The wave is sent into a Michelson interferometer; one arm is lengthened relative to the other so as to introduce a time delay τ between the recombining fields, which have equal intensities I_0 . Show that the detected intensity may be written

$$I(\tau) = 2I_0 [1 + |g(\tau)| \cos \varphi(\tau)],$$

identifying $\varphi(\tau)$. Sketch and annotate $I(\tau)$ for (i) a perfectly monochromatic wave and (ii) a quasi-monochromatic wave, and explain how the interferogram allows $|g(\tau)|$ to be measured. [5]

- (c) A light-emitting diode has a centre wavelength of $\lambda_0 = 800$ nm and a spectral width of $\Delta\lambda = 25$ nm. Estimate its coherence length. Explain why a short coherence length is desirable for an interferometric imaging technique that profiles a multilayered sample. [4]
- (d) Optical coherence tomography (OCT) uses such a source to image a sample consisting of several weakly reflecting boundaries at depths giving delays τ_i . Explain qualitatively why each boundary produces a distinct burst of interference fringes in the interferogram, and state how the axial separation of two layers is extracted from the measurement. [4]
- (e) State one reason why the axial resolution of OCT is, unlike that of a conventional microscope, independent of the numerical aperture of the imaging optics. [4]

6 This question concerns the quantised single-mode electromagnetic field and the states of light it supports.

A single mode of the electromagnetic field is described by the Hamiltonian $\hat{H} = \hbar\omega(\hat{a}^\dagger\hat{a} + \frac{1}{2})$, where $\hat{a}$ and $\hat{a}^\dagger$ obey $[\hat{a}, \hat{a}^\dagger] = 1$. The electric field operator may be taken as $\hat{E}_x \propto (\hat{a} + \hat{a}^\dagger)$.

- (a) The number (Fock) states $|n\rangle$ are the eigenstates of $\hat{n} = \hat{a}^\dagger\hat{a}$. Using $\hat{a}|n\rangle = \sqrt{n}|n-1\rangle$ and $\hat{a}^\dagger|n\rangle = \sqrt{n+1}|n+1\rangle$, show that $\langle n|\hat{E}_x|n\rangle = 0$ for all n , while $\langle n|\hat{E}_x^2|n\rangle \propto (n + \frac{1}{2})$. Comment on why a Fock state, even with large n , does not behave like a classical light field. **[5]**

- (b) A *coherent state* $|\alpha\rangle$ is defined as the eigenstate of the annihilation operator, $\hat{a}|\alpha\rangle = \alpha|\alpha\rangle$ with $\alpha \in \mathbb{C}$. Given that

$$|\alpha\rangle = e^{-|\alpha|^2/2} \sum_{n=0}^{\infty} \frac{\alpha^n}{\sqrt{n!}} |n\rangle,$$

show that the probability of detecting n photons is Poissonian, and hence that the mean photon number is $\bar{n} = |\alpha|^2$ and the variance is $(\Delta n)^2 = \bar{n}$. **[5]**

- (c) Explain why a coherent state, unlike a Fock state, has a non-zero expectation value of the field, and state how the fractional uncertainty in the photon number, $\Delta n/\bar{n}$, behaves as $|\alpha| \rightarrow \infty$. Why does this justify regarding the coherent state as the quantum description of an ideal laser beam? **[4]**

- (d) A single photon, in the state $|0\rangle_0 |1\rangle_1$, is incident on one input port of a lossless 50:50 beam splitter, the other input port being in the vacuum state. Using the beam-splitter relation $\hat{a}_1^\dagger = \frac{1}{\sqrt{2}}(i\hat{a}_2^\dagger + \hat{a}_3^\dagger)$, find the output state. Comment on the role played by the vacuum entering the unused port. **[3]**

- (e) Two indistinguishable single photons are now incident, one on each input port of the same beam splitter. State, with a brief justification, what is observed in a coincidence measurement at the two outputs, and explain how this effect is used to test whether two photons are truly indistinguishable. **[3]**

END OF PAPER